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- **The theme (design) of this template must be used for your pitch slides.**
- **All slides must be written in English. Presenting in English is strongly encouraged.**
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- Grey text in [square brackets] is guidance only. Replace it with your own content.
- Your team decides how to split the slot between presentation and demo.
- Save as PDF or PPTX and submit before Day 3 at 08:00 WIB.

T01 – NaS

SeBatt

Secure Battery Identity & Tamper-Evident History

Internet of Things · Universitas Gadjah Mada

Guntur Sulistyo · Muhammad Budi Setiawan · Muhammad Muqtada Alhaddad

Hackathon Deeptech 2026 · Demo Day, 23 October 2026



Condition

139,45 M
Motorcyles
+ 21% since 2020

167.864
2 Wheel EV
+ 50X since 2020

2,240
Battery Swap Station
+ 6X since 2022

Problem

Battery are shared, so the station must trust a battery.



Originality



Integrity Data
– Manipulation
– Loss or Reset

Solution

SeBatt
SECURITY BATTERY

Unique Chip Identifier

Signature on every data

Monotonic Sequence Number

Key Activities

IP License Sales
B2B selling to battery and BMS makers, from pilot to signed license.

After-Sales Support
Integration help, security updates, keeping each licensee in production.

Identify Registration
Give every chip a unique identity and record it when the battery is made.

Battery Verification
Check signature and sequence number each time a battery enters a station.

Key Resource

IP Design & HKI
RTL with HMAC-SHA-256 and sequence counter; patent/ HKI to protect it.

IC Design & Security Team
Hardware design, verification, and cryptography skills in one team.

Battery Identity Registry
Secure database of chip certification run with an HSM for key safety.

R&D Certification Funfs
Tape-out and security certification are the biggest one time costs.

Key Partnership

Regulator
Sets verification as a standard and creates pull demand from operators.

Security Test Lab
Independet certification that buyers and regulators trust.

Foundry & MPW
First silicon on a shared wafer far cheaper than a full mask set.

Customer Segment

Battery & BMS Makers
Pays license + royalty
Build the chip into every new swap battery

Fleets with their own BSS
Pays verification service
Own batteries and stations, so misuse hit them directly

Open Swap Operators
Pays nothing to us
Accept only batteries carrying a verified chip.

Channels

Awareness
ICDeC demo day, technical write-up, EV and battery industry events

Evaluation
FPGA demo board + pilot at a fleet's own stations

Purchase
Direct B2B sales like license contract signed with milestones.

Delivery
IP Package + reference design + integration engineer on site.

After Sales
Registry access, verification tools, security update.

Customer Relationship

Dedicated Assistance
aAn integration engineer per licensee, drom first RTL drop to production.

Co-creation
Pilot with the fleet, reference design shaped by their feedback.

Long Term Contract
License + royalty for the product life, not a one off sale.

Ongoing Security
Registry, Key Management, and patches keep the chip trusted.

Cost Structure

One Time
Security Certification
MPW Tape Out x2
Registry Server + HSM
Lab Tools, HKI, PCB, Fleet Pilots

Recurring
Salary
Office & Operation
Marketing & Events
EDA Tools, Server, Legal

Variable Cost
Sales Cost
Integration Engineer
Registry per Battery

Revenue Stream

Licence Fee
Build the chip into every new swap battery

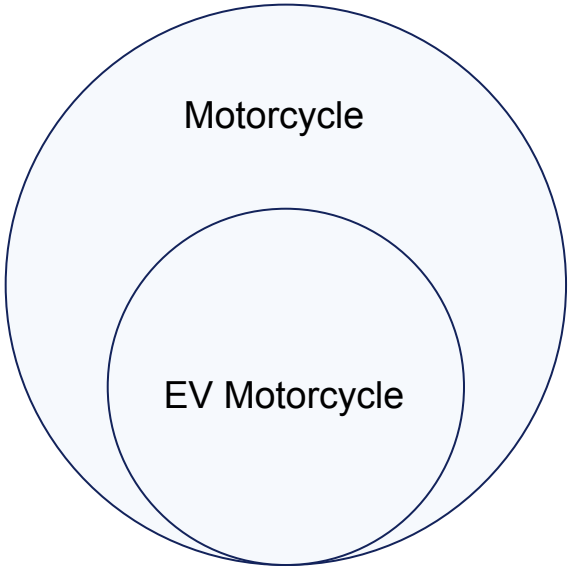
Integration Service
Own batteries and stations, so misuse hit them directly

Verification Service
Accept only batteries carrying a verified chip.

Royalty
Accept only batteries carrying a verified chip.

Trend of EV Motorcycle

Who has the problem, how big it is, and the evidence



- Based on BPS 2024, Indonesia had 139,45 Million Unit Motorcycle. It is **increased around 17%** in 4 years.
- EV Motorcycle in indonesia in that year is 167.684 unit. **It is increased around 4,494%** in 5 years.
- SPBKLU as a BSS **had increased around 510%** in 2 years.

Battery Swap EV Motorcycle

Who has the problem, how big it is, and the evidence

Battery swap is one of kind mechanism for **recharge the battery of ev motorcycle**. Driver do not need to waiting battery recharge, it can be **replace by fully recharge battery** in battery swap station.



Originality



Integrity Data



How SeBatt can Solve the Problem

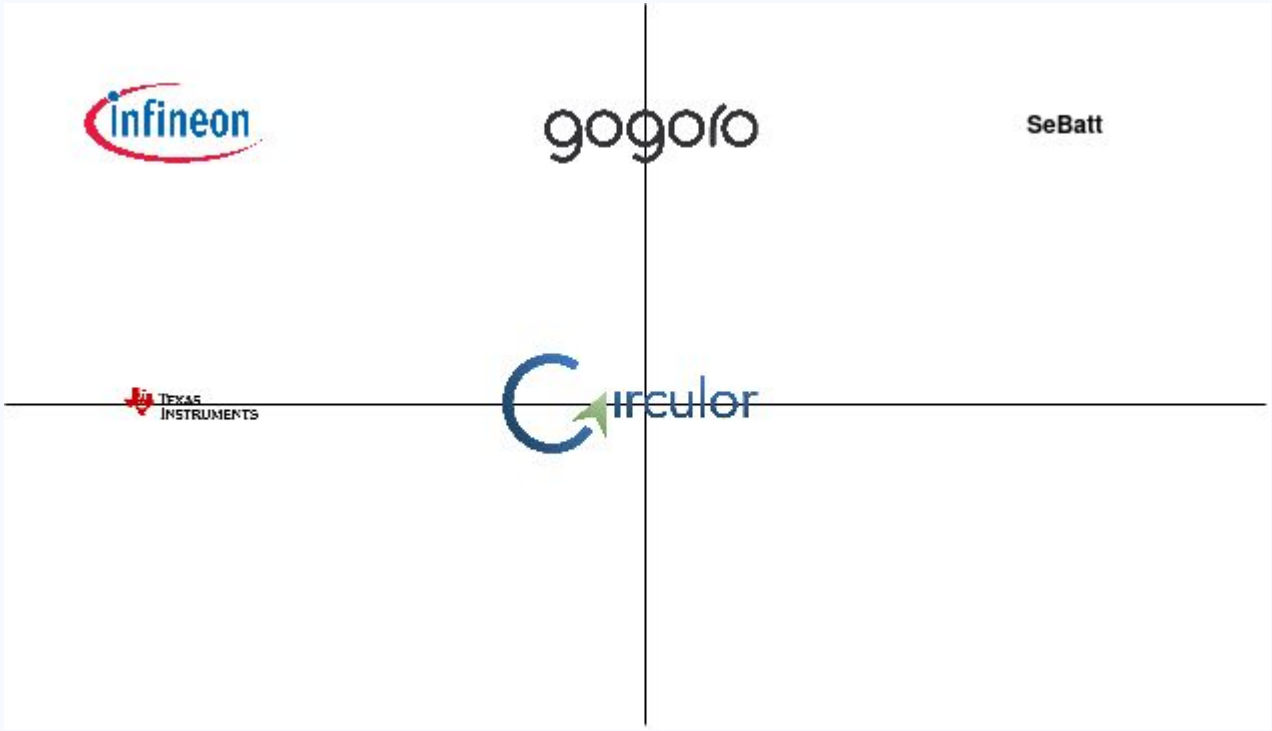
Who has the problem, how big it is, and the evidence

Problem	SeBatt Feature	Output
Originality	Unique key per chip	Counterfeit or unregistered batteries are rejected by the station
Data Integrity - Data Modification	HMAC signature for every sample	Modified data fails verification
Data Integrity - Removed Data or Reset	Monotonically increasing sequence number	Missing samples or outdated data are detected

Positioning SeBatt

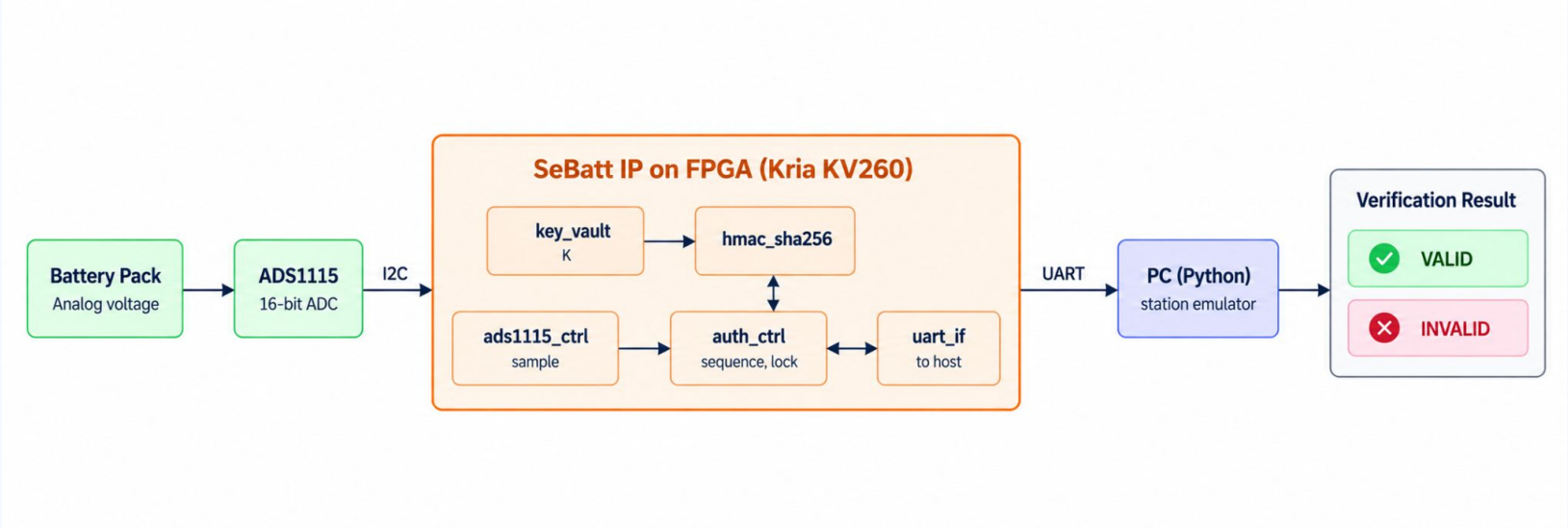
Kemampuan	DS28E38	BQ40Z50	ADI wBMS	SeBatt
Proves battery authenticity	Ya (ECC, PUF)	Ya, (HMAC-SHA-1)	Tidak jelas	Ya (HMAC-SHA-256)
Measures voltage on the same chip	Tidak	Ya	Ya, melalui cell monitor	Ya, target ASIC
Merekam riwayat di dalam chip	Tidak	Ya	Tidak jelas	Ya
Riwayat ditandatangani, penghapusan dan rollback terdeteksi	Tidak	Tidak, hanya kata sandi	Tidak terdokumentasi	Ya
Diverifikasi pihak di luar baterai	Identitas	Identitas	Tidak, internal	Identitas dan

Competitor Analysis



Solution and system architecture

How it works from input to output



The measurement is authenticated on the hardware side; a verifier checks the cryptographic result and reports whether the tested record matches. The current PoC validates fixed-input RTL and offline verification; live ADC-to-history integration is still planned.

Role of the chip

Which block you designed and why a dedicated chip is needed

1

Designed block

SHA-256 / HMAC RTL, AUTH control, UART interface,

2

Why a chip

On Hardware Authentication will make it harder to flashed / manipulated

3

Implementation

Both chip and station will have key to authenticate their history and the stored data key calculating hash in match

PoC demo and evaluation results

What works and how well it performs

[Photo of the demo setup, screenshot, or waveform]

Metric	Result	Target / baseline
[Accuracy]	[...]	[...]
[Latency]	[...]	[...]
[Throughput]	[...]	[...]
[Power / FPGA resources]	[...]	[...]

[If the input comes from data rather than a live sensor, state it here]

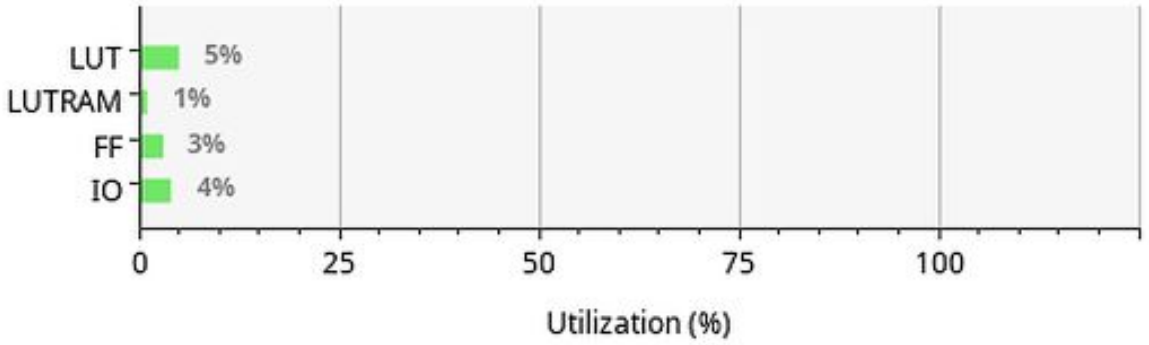
Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	0.675 ns	Worst Hold Slack (WHS):	0.011 ns	Worst Pulse Width Slack (WPWS):	4.238 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	18495	Total Number of Endpoints:	18495	Total Number of Endpoints:	6574

All user specified timing constraints are met.

Utilization

Resource	Utilization	Available	Utilization %
LUT	5614	117120	4.79
LUTRAM	376	57600	0.65
FF	6198	234240	2.65
IO	7	189	3.70

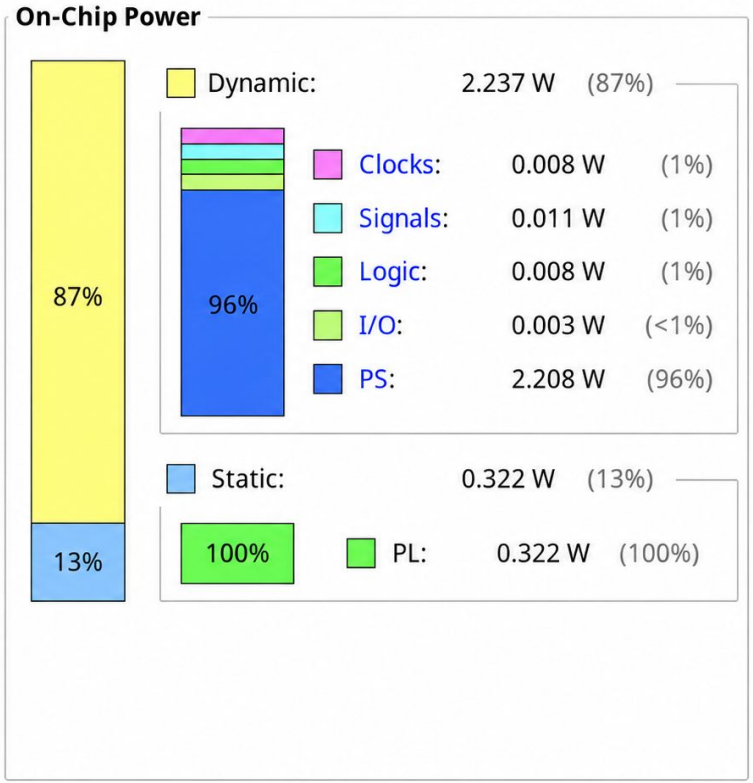


Power analysis

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	**2.559 W**
Design Power Budget:	**Not Specified**
Process:	typical
Power Budget Margin:	**N/A**
Junction Temperature:	**30.9°C**
Thermal Margin:	54.1°C (23.0 W)
Ambient Temperature:	25.0 °C
Effective θJA:	2.3°°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Low

Launch Power Constraint Advisor to find and fix invalid switching activity



Business model

Business Model Canvas summary

Key Partnerships
– Foundry & MPW Program
– Lembaga uji keamanan

Key Activities
– IP development
– Integrasi ke produsen
– Registrasi Baterai
– Operasi Verifikasi

Key Resources
– Desain IP & HKI
– Tim Desain IC & Keamanan
– Registri identitas baterai
– Dana R&D dan Sertifikasi

Value Propositions
Riwayat pemakaian baterai yang bisa dibuktikan keasliannya, bukan sekedar dilaporkan.

Customer Relationships
– Pendampingan integrasi IP
– Kontrak Jangka Panjang
– Pilot bersama armada
– Dashboard Riwayat Baterai

Channels
– Lisensi langsung ke produsen
– Permintaan operator
– Pameran dan asosiasi EV

Customer Segments
– Produsen baterai & BMS (pembeli lisensi)
– Armada dengan BSS sensiri (pengguna verifikasi)
– Operator swap terbuka (pengguna verifikasi)

Cost Structure
– R&D Desain IP
– Prototipe Silikon (MPW)
– Sertifikasi Keamanan
– Gaji Tim
– Server dan Keamanan Kunci
– Support after sales B2B

Revenue Streams
– Lisensi IP per desain
– Royalti per chip
– Layanan verfikasi
– Jasa integrasi

Our business model

Business Model Canvas summary

Key Partnerships
– Foundry & MPW Program
– Security Lab

Key Activities
– IP development
– Integration to manufacturers
– Battery Registration
– Verification Operations

Key Resources
– IP Design & IP Rights
– IC Design & Security Team
– Battery Identity Registry
– R&D and Certification Funding

Value Propositions
Verifiable battery usage history that proves authenticity, rather than relying solely on reported data.

Customer Relationships
– IP Integration Support
– Long-Term Contracts
– Joint Pilots with Fleets
– Battery History Dashboard

Channels
– Direct IP Licensing
– Operator Procurement
– EV Exhibitions and Associations

Customer Segments
– Battery & BMS Manufacturers (License Buyers)

– Fleets with Their Own BSS (Verification Users)

– Open Battery-Swapping Operators (Verification Users)

Cost Structure
– IP Design R&D
– Security Certification
– Server and Key Security
– Prototipe Silikon (MPW)
– Sertifikasi Keamanan
– Silicon Prototyping (MPW)
– Team Salaries
– B2B After-Sales Support
– Server dan Keamanan Kunci
– Support after sales B2B

Revenue Streams
– Lisensi IP per desain
– Layanan verfikasi
– Royalti per chip
– Jasa integrasi

Development roadmap

1 — 2 — 3

Design Phase

– Build the PoC
– Interview fleets, swap operators, and battery makers

Alpha Phase

– Implement signatures mechanism and non volatile history data
– Develop station verification software.

Beta Phase

– Pilot in one fleet with its own swap stations
– Onboard a battery or BMS maker as pilor partner
– Tape out the first test chip through an MPW program
– Secure letters of intent for licensing

Post Release

– Obtain security certification
Sign the first IP license and support integration
– Chip enters mass production.

Thank you

T01 – NaS

[Team contact or repository link]



Business model

Business Model Canvas summary

Key Partnerships
– Foundry & Program MPW
– Lembaga uji keamanan

Key Activities
– Pengembangan IP
– Integrasi ke produsen
– Registrasi Baterai
– Operasi Verifikasi

Key Resources
– Desain IP
– Tim Desain IC & Keamanan
– Basis data & Server kunci
– Dana R&D Prototipe

Value Propositions
Riwayat pemakaian baterai yang bisa dibuktikan keasliannya, bukan sekedar dilaporkan.

Customer Relationships
– Pilot bersama operator
– Pendampingan Integrasi
– Kontrak Jangka Panjang
– Dashboard Riwayat Baterai

Channels
– Penjualan ke operator
– Lisensi langsung ke produsen
– Pameran dan asosiasi EV

Customer Segments
– Operator Baterai Swap
– Produsen baterai dan BMS
– Armada EV dengan BSS sendiri

Cost Structure
– R&D Desain IP
– Prototipe Silikon (MPW)
– Sertifikasi Keamanan
– Setup server dan HSM
– Gaji Tim
– Server dan Keamanan Kunci
– Support after sales B2B

Revenue Streams
– Subcription Verifikasi
– Lisensi IP per desain
– Royalti IP per chip

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[Solution Title]

[One sentence: what problem you solve and for whom]

Internet of Things · Universitas Gadjah Mada

[Member 1] · [Member 2] · [Member 3] · [Member 4] · [Member 5]

Hackathon Deeptech 2026 · Demo Day, 23 October 2026



Problem and target users

Who has the problem, how big it is, and the evidence

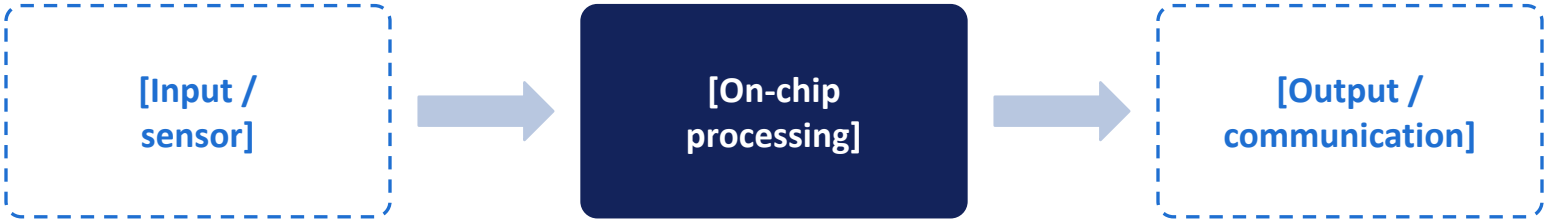
- [The main problem you address]
- [Who your target users are]
- [Impact if the problem is not solved]
- [Limitations of current solutions]

[Key figure]

[Short explanation of the figure and its data source]

Solution and system architecture

How it works from input to output



[Replace with your system block diagram]

[Short description of the system workflow, 2–3 sentences]

Role of the chip

Which block you designed and why a dedicated chip is needed

1

Designed block

[Accelerator / IP / RTL module built by the team]

2

Why a chip

[Reason: power, latency, size, cost, or privacy]

3

Implementation

[RTL, KV260 FPGA, simulation, or ASIC flow used]

PoC demo and evaluation results

What works and how well it performs

[Photo of the demo setup, screenshot, or waveform]

Metric	Result	Target / baseline
[Accuracy]	[...]	[...]
[Latency]	[...]	[...]
[Throughput]	[...]	[...]
[Power / FPGA resources]	[...]	[...]

[If the input comes from data rather than a live sensor, state it here]

Business model

Business Model Canvas summary

Key Partnerships

[...]

Key Activities

[...]

Key Resources

[...]

Value Propositions

[...]

Customer Relationships

[...]

Channels

[...]

Customer Segments

[...]

Cost Structure

[...]

Revenue Streams

[...]

Development roadmap

Next steps after the hackathon

1

Short term

[PoC refinement, further testing]

2

Mid term

[Prototype, user testing, industry partners]

3

Long term

[Tape-out, production, commercialization]

Thank you

[Team ID] – [Team name]

[Team contact or repository link]

